Quick Access Recorder (QAR) (S.O.O. 6356) – General Data

1. General

The Quick Access Recorder (QAR) Installation (S.O.O. 6356) consists of a removable QAR unit which is wired to the FDR DATAOUT channels. The QAR continuously records data from the aircraft Flight Data Recorder when the QAR switch is in the NORMAL position. The data is subsequently downloaded to a Personal Computer (PC) for analysis.

2. Description

The QAR is a data capture and storing system. The QAR unit provides multiple inputs (digital and analogue) for gathering data from onboard sensory equipment. The QAR comes with a 2 gigabyte (GB) removable CompactFlash Card, which will provide 1280 hours of recording time at 256 Words-Per-Second (WPS) data stream. The data from the QAR may be downloaded onto a computer while installed in the aircraft, using the built-in mini Universal Serial Bus (USB) port. In addition, the USB port may be used to supply power to the QAR from a host computer when aircraft power is not available.

The QAR is located above the rear baggage door on the left hand side of the aircraft at station 362.00. The QAR circuit breaker (T3), located on the station 332 circuit breaker panel, is powered through the left bus.

3. Equipment and Materials

Optional (required to download data directly from QAR via the USB port):

- Personal Computer (PC) with Microsoft Windows, standard USB port interface (ver. 1.1 or 2.0) or PCMCIA drive
- microQAR Flight Data Processor software utility program (L-3 Communications part number 883E2822–03 or above).
- USB cable with ‘standard’ connector and ‘mini-B’ connector.
- L-3AR Rose software.
- Database part number 851–E6415–00.

4. Adjustment/Test

A. Function Test of the QAR

1) Gain access to the QAR in the rear baggage compartment, above the baggage door.

2) Remove QAR cover per Para 5.A.4).

3) Set QAR switch to the OFF position.

4) Switch aircraft power ON.

5) Ensure that both LEDs on the QAR are not illuminated.

6) Set QAR switch to the NORMAL position. Ensure that both the LEDs on the QAR momentarily illuminate two times and then go out; this indicates normal operation.

7) Set QAR switch to the OFF position.

8) Open the QAR (T3) (Left Bus) circuit breaker on the stations 332 circuit breaker panel.

9) Set QAR switch to the NORMAL position. Ensure that both LEDs on the QAR are not illuminated.

10) Close the QAR (T3) (Left Bus) circuit breaker on the station 332 circuit breaker panel. Ensure both LEDs located on the label side of the QAR momentarily illuminate two times and then go out; this indicates normal operation.

11) Install QAR cover per Para 5.B.5).

5. Removal/Installation

A. Removal

1) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

2) Open the QAR (T3) (Left Bus) circuit breaker on the station 332 avionics circuit breaker panel and fit with safety clip.

3) Gain access to the QAR in the rear baggage compartment, above the baggage door.

4) Remove the two (2) screws securing the QAR cover to the airframe.

5) Remove the QAR by disconnecting mating connector.

Note Removal of airframe mounted connector is not required.

B. Installation

1) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

2) Ensure the QAR (T3) (Left Bus) circuit breaker on the station 332 avionics circuit breaker panel is open and fitted with safety clip.

3) Gain access to the QAR installation area in the rear baggage compartment, above the baggage door.

4) Install QAR to mating connector.

5) Install the two (2) screws and washers securing the QAR cover to the airframe.

6) Close the QAR (T3) (Left Bus) circuit breaker on the station 332 circuit breaker panel.

7) Do a function test of the QAR, per Para 4.A.

6. Troubleshooting

A. QAR Unit

1) If the QAR is deemed inoperative check the following:

a) Ensure that the QAR (T3) (Left Bus) circuit breaker is closed.

b) Ensure that the QAR switch is in the NORMAL position.

c) Check the CompactFlash card is not full.

d) Check that the QAR is properly secure and grounded to the airframe.

e) Ensure that all the connectors are mated and tight.

f) Visually inspect the QAR for damaged wires and contacts, and ensure no short circuits have formed.

g) If troubleshooting at a wire level is required, refer to the applicable diagrams in the Wiring Diagrams Manual PSM 1-64-2W.

h) If repair or service is required, return the QAR to L-3 Communications.

2) The LED indicators on the QAR may assist in problem diagnosis:

FAULT LED	QAR STATUS
LED On – Amber	QAR has stopped functioning and should be removed from aircraft and returned to L-3 Communications.

MEMORY FULL LED	
LED On – Amber	Memory card is full and recording will stop. CompactFlash card may be removed and replaced or the data from the card may be downloaded and cleared at the aircraft thorough the QAR micro USB interface.

MEMORY FULL AND FAULT LED	
LEDs momentarily flash two times on start up	QAR operating normally.

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